

Spirit in Physics:

A falsifiable physical hypothesis of spirit, pilot feasibility study, and prospective validation framework

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Abstract

Spirit in Physics advances the working hypothesis that spirit is a physically realized, information-bearing organization whose structure can be measured. Participant-specific response manifolds are proposed as an operational approximation; response energy, Jungian complexes, and a component corresponding to the collective unconscious are explicit test targets. Stage 1 asks whether repeated word-association measurements can be converted into auditable geometries. Eleven participant directories were present; five contributed 970 valid clock-aligned trials over two sessions and 99 shared stimulus words. Test–retest latency correlation was weak ($r = 0.24$ across 471 matched participant–word pairs). Mean cross-participant latency correlation was small ($\bar{r} = 0.065$; two-sided permutation $p = 0.038$), whereas the first-PC share (27%) did not exceed its shuffle null (approximately 26%, $p = 0.21$). A rank-[3, 6, 3] tensor approximation described 75.9% of variance. We therefore pair the feasibility result with a prospective Stage 2 framework: H_{phys} is primary; H_{complex} and $H_{\text{collective}}$ are key secondary hypotheses; and H_{energy} is a mechanistic long-term hypothesis requiring calibrated physical quantities. The framework prespecifies independent replication, held-out prediction, controlled perturbation, external construct validation, negative controls, and falsification conditions. Stage 1 supports construction feasibility, not proof of the four hypotheses.

1 Introduction

Reaction time and autonomic activity during word association have long been used to study stimulus-linked variation in human responses [1]. Modern graph and tensor methods provide compact ways to summarize such multivariate observations [2]. They do not, by themselves, establish a psychological construct or causal mechanism.

Information theory and the physical cost of information processing motivate the project’s physical premise [3–5]. The long-term program asks whether the informational organization called spirit has a measurable physical realization. Analytical psychology and the rubber-hand illusion [6] provide candidate structures and bodily-boundary perturbations through which that premise can be tested.

The objective of the program is therefore stronger than visualization: to accumulate reproducible, intervention-sensitive evidence capable of supporting or falsifying a physical model of spirit. The objective of this first installment is narrower: to determine whether latency and skin-potential recordings can be transformed into an auditable candidate representation and to identify which parts of the hypothesis remain untested.

2 Working premise and testable hypotheses

We adopt the following premise: a person’s spirit is physically realized as an information-bearing organization of bodily and cognitive states. The measured manifold is not declared identical to that organization; it is a candidate model whose adequacy must be tested by prediction, perturbation, and replication. For Stage 2, H_{phys} is the primary hypothesis, H_{complex} and $H_{\text{collective}}$ are key secondary hypotheses, and H_{energy} is mechanistic and long-term until a calibrated physical quantity is available.

H_{phys} : If spirit has a physical realization, its estimated state should reproducibly predict held-out behavioral and physiological responses and should change systematically under controlled perturbations of bodily self and information processing.

H_{energy} : Regions requiring greater response latency and autonomic change are observable proxies for higher response cost. A physical-energy interpretation additionally requires calibrated measurements linking the proxy to energetic or information-thermodynamic quantities; the present score alone is insufficient.

H_{complex} : Stable high-cost regions should correspond to independently assessed, personally salient Jungian complexes and predict response content or physiology beyond the variables used to construct the regions.

$H_{\text{collective}}$: After individual and technical effects are controlled, a replicable cross-participant component should remain and should generalize across samples. Its relation to Jung’s collective unconscious is supported only if it also shows theoretically predicted semantic or cultural structure.

Failure of out-of-sample prediction, perturbation sensitivity, reliability, or independent construct validation would count against the corresponding hypothesis. The current pilot principally addresses whether the candidate representation can be constructed; it provides only exploratory evidence relevant to the remaining hypotheses.

3 Stage 1 pilot methods

3.1 Ethics and consent

Project materials report approval by Niigata University (approval 2024-0269; March 1, 2025). The study interface required acknowledgment of the research purpose, voluntary participation, the right to withdraw without disadvantage, and audio/video recording before registration. The approval letter and retained participant-level consent records were not part of the reviewed analysis package; their completeness must be confirmed before submission.

3.2 Participants and analytic set

Eleven participant directories were present. Six could not enter the complete-record analysis: three continuous-recording files were unavailable git-annex pointers, two directories contained no physiological CSV, and one recording did not overlap the task-event clock. Five participants contributed analyzable synchronized records. This was an availability-based analytic subset, not a random or prospectively selected sample. No demographic or clinical inference is attempted.

3.3 Task and signal extraction

Participants completed two presentations of an approximately 100-word stimulus list. The present analysis uses stimulus identity, event timing, and continuous skin-potential signals; semantic content of spoken association responses is not used in the manifold.

Continuous Mod-002 CSV recordings were sampled at 1 Hz. Response latency was the interval from `response_window_opened` to the first `speech_detected` event; values had to

satisfy $0 < \text{latency} < 12$ s. For each channel, skin-potential change ΔSP was the maximum absolute deviation during the response window from the mean of the 3 s preceding stimulus display. A trial was retained only when event times fell inside the recording and all four channel windows were available. This yielded 970 valid trials (198, 181, 195, 199, and 197 by participant).

Channel 2 is the primary skin-potential channel in the current analysis code; Channel 1 is retained as an additional tensor feature. This choice was not preregistered, and electrode-placement metadata sufficient to give the channels a stable physiological interpretation were unavailable.

3.4 Participant-specific response spaces

Within each participant, trials were averaged by stimulus word. The three features were latency, ΔSP_{Ch1} , and ΔSP_{Ch2} . For visualization we define a descriptive response-cost score

$$C(w) = z\{\text{latency}(w)\} + z\{\Delta SP_{Ch2}(w)\}. \quad (1)$$

This dimensionless score is a candidate proxy under H_{energy} , not a direct measurement of physical energy. Its validation requires independent energetic or information-thermodynamic measurements.

After feature standardization, squared Euclidean distances generated the radial-basis affinity

$$W_{ij} = \exp\{-\|x_i - x_j\|^2/\sigma^2\}, \quad (2)$$

where σ was the median nonzero pairwise distance. Coordinates were the nontrivial low-order eigenvectors of the symmetric normalized graph Laplacian $L_{\text{sym}} = I - D^{-1/2}WD^{-1/2}$. This is a graph-spectral response embedding, not Fisher information geometry.

3.5 Cohort tensor and exploratory statistics

Participant-by-word means were assembled as $\mathcal{T} \in \mathbb{R}^{P \times W \times F}$ with $P = 5$, $W = 99$, and $F = 3$. Missing cells were imputed first by the word-feature mean and then by the feature mean. Each feature was standardized across the participant-word plane. A truncated HOSVD/Tucker approximation [2] used ranks [3, 6, 3]. The graph spectrum, participation ratio, and reconstructed variance are descriptive summaries whose ranks were not selected on an independent sample.

Test-retest stability was Pearson correlation of session-1 and session-2 latency over available matched participant-word pairs. Cross-participant structure was evaluated from the participant-by-word latency matrix. The mean off-diagonal Pearson correlation and first-principal-component variance share were compared with null distributions from 2,000 within-participant word permutations using seed 7. The correlation test was two-sided. These analyses were exploratory, were not preregistered, and no multiplicity-adjusted confirmatory claim is made.

4 Prospective Stage 2 validation framework

The next large-dataset study is planned as a prospective validation stage. This section is a framework, not a completed registration. Before confirmatory data are inspected, the protocol, code version, endpoint definitions, effect directions, success thresholds, exclusions, missing-data rules, and analysis order should be timestamped in a public registry.

4.1 Design, cohorts, and sample size

Stage 2 will use repeated measurements sufficient to estimate reliability and held-out prediction. Sample size will be determined by simulation across plausible reliability, attrition, clustering,

and effect-size ranges rather than by treating the unstable Stage 1 estimates as true effects. A discovery cohort will be separated from an independent replication cohort; where cultural generalization is claimed, an additional language or cultural cohort is required. Participant assignment, intervention order, device version, sensor placement, calibration, and acquisition failures will be retained in a machine-readable flow record.

4.2 Prospective decision table

Table 1: **Prospective validation logic.** Exact endpoints and thresholds will be frozen before confirmatory data inspection.

Hypothesis	Primary validation	Required comparison	Evidence against the hypothesis
H_{phys}	Held-out behavioral/physiological prediction and a prespecified bodily-self perturbation	Participant-agnostic, lexical, sham, and unperturbed baselines	No incremental prediction or reproducible intervention-sensitive change
H_{energy}	Association of C with a separately calibrated energetic or information-thermodynamic quantity	Latency-only, arousal-only, and non-energetic nuisance models	No relation to an independent physical quantity after confound control
H_{complex}	Stable high- C regions predict blinded, independently measured personal salience or complex criteria	Word frequency, valence, latency, and skin-potential components	No reliability or incremental external prediction
$H_{\text{collective}}$	Prespecified shared semantic structure replicates in independent and cross-cultural samples	Shuffle, lexical-frequency, language, device, site, and demographic baselines	The component disappears after controls or fails external replication

4.3 Analysis hierarchy and leakage control

Reliability is the first gate: an unstable representation is not advanced to strong construct claims. The primary H_{phys} analysis will use participant-level separation of training, validation, and test data, with preprocessing fitted inside the training partition. Baseline comparisons and negative controls will be evaluated on the same held-out observations. Hierarchical models will account for participant, word, session, site, and device structure as appropriate. The key secondary hypotheses will be tested only after the primary analysis plan is executed; multiplicity control, directional contrasts, and any Bayesian decision criteria will be fixed in the registration. Exploratory analyses will be labelled and will not redefine success.

4.4 Evidence levels

We define a cumulative evidence ladder: Level 0, constructibility (the present pilot); Level 1, test-retest reliability; Level 2, held-out prediction; Level 3, response to controlled perturbation; Level 4, independent energetic and complex construct validity; and Level 5, independent and cross-cultural replication of shared structure. Support for a physical model of spirit requires at least Level 3. A collective-unconscious interpretation requires Level 5 and prespecified semantic or cultural structure, not principal-component variance alone.

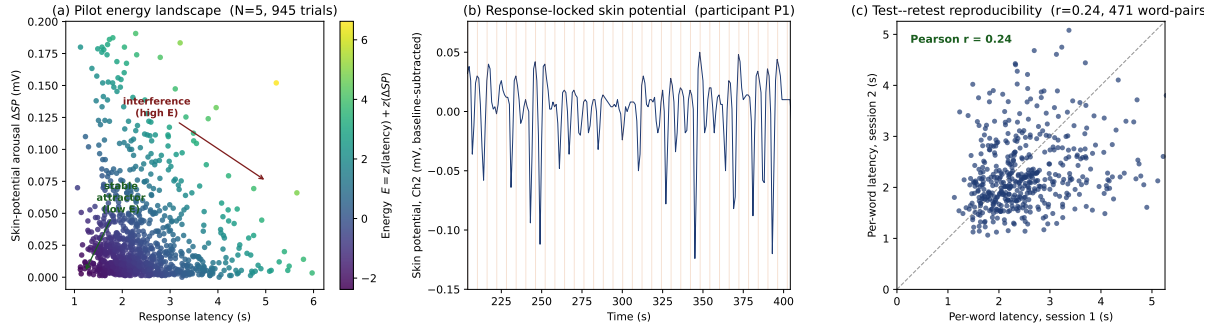


Figure 1: **Pilot measurements.** (a) Response latency and Channel-2 skin-potential change for 970 valid trials, colored by the descriptive response-cost score C . (b) Example continuous skin-potential trace and task events. (c) Session-1 versus session-2 latency for matched participant–word observations.

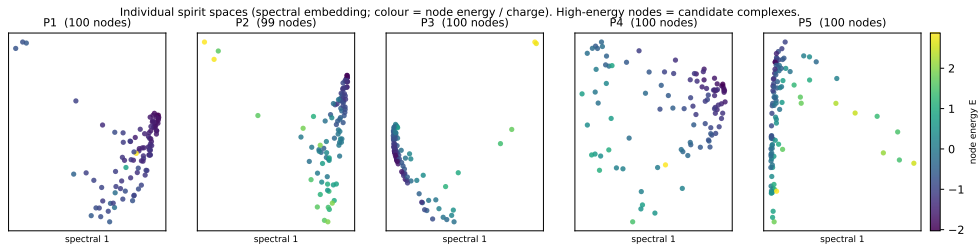


Figure 2: **Participant-specific response spaces.** Each panel is a graph-spectral embedding of one participant’s word nodes from latency and two skin-potential features; color is C . Axes have arbitrary sign and orientation and are not directly comparable without alignment.

4.5 Preregistration and release

Before enrollment or confirmatory data access, Stage 2 should be preregistered with a frozen analysis commit. The public package should contain a versioned environment, checksums, synthetic or ethics-compatible derived test data, participant flow, endpoint dictionary, and a one-command reproduction path. Deidentified derived results and code should receive a durable archive identifier; participant-level media remain governed by consent and ethics approval.

5 Stage 1 pilot results

5.1 Trial-level response landscape

All 970 valid trials are shown in Fig. 1; no percentile trimming is used in the revised display. Because color is computed from the plotted axes, separation of low and high C is expected by construction. It demonstrates measurement coverage, not latent classes. Session-specific latency stability was weak ($r = 0.24$, 471 matched participant–word observations).

5.2 Participant-specific embeddings

The same algorithm produced a two-dimensional response embedding for each participant (Fig. 2). Under H_{complex} , stable high- C regions are candidate complex markers. Visual differences are descriptive, and no independent complex assessment or validated classifier is available in this pilot.

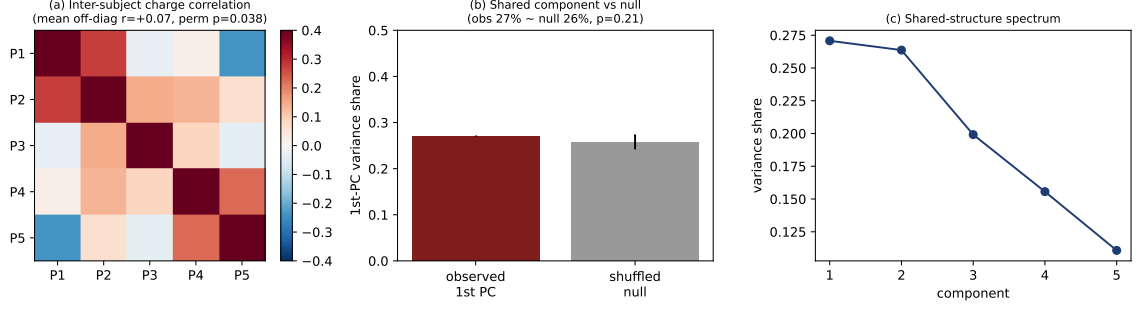


Figure 3: **Exploratory cross-participant latency analysis.** (a) Pairwise correlations and their mean, with a two-sided within-participant permutation test. (b) Observed first-PC variance share versus the shuffle null. (c) Latency variance spectrum. These panels analyze latency, not a physiological “charge.”

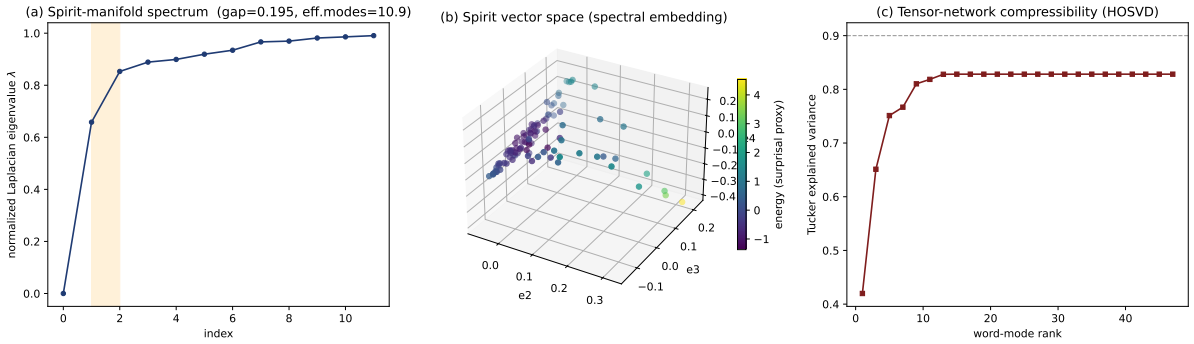


Figure 4: **Descriptive cohort geometry.** Graph-Laplacian spectrum, three-dimensional word embedding, and Tucker reconstruction curve for the $5 \times 99 \times 3$ tensor. The summaries are in-sample and descriptive.

5.3 Cross-participant latency and tensor summaries

Mean off-diagonal cross-participant latency correlation was $\bar{r} = 0.065$ and exceeded the absolute shuffled statistic in the two-sided permutation test ($p = 0.038$). The first principal component explained 27%, close to the approximately 26% shuffle-null mean ($p = 0.21$; Fig. 3). This is a faint exploratory observation relevant to $H_{\text{collective}}$, but the principal-component result provides no evidence for a large common component and the semantic criteria of the hypothesis were not tested.

The tensor had shape $5 \times 99 \times 3$. The median-distance kernel scale was $\sigma = 4.2109$, the graph eigengap was 0.1948, and the spectrum’s participation ratio was 10.89. Rank-[3, 6, 3] HOSVD reconstructed 75.9% of variance (Fig. 4). With only five participants and three features, these values partly reflect dimensional constraints and do not establish a population manifold or a unique “collective coordinate.”

6 Discussion

The four hypotheses are the scientific center of this research program, rather than decorative interpretations added after analysis. Stage 1 establishes the first bounded result needed to pursue them: synchronized word-association and skin-potential data can be transformed into participant-specific graph embeddings and a cohort tensor through an auditable computation. Stage 2 converts the program from a broad aspiration into a prospective sequence of reliability,

prediction, perturbation, independent validation, and replication tests. It does not predetermine their outcome.

The physical premise becomes scientific only through discriminating measurements. In the present installment no temperature, heat, work, erasure event, or entropy production was measured, so C is an energy hypothesis proxy rather than thermodynamic energy. Similarly, the radial-basis graph is a first operational geometry, not yet a Fisher-metric statistical manifold. Later installments must link the candidate space to calibrated energetic quantities and controlled interventions if H_{phys} and H_{energy} are to receive support.

The weak test–retest correlation indicates that measurement noise, learning, state change, or event alignment may be substantial. The small permutation result is hypothesis-generating: it was derived after inspection of a small complete-case subset, without multiplicity control, and the PC analysis was null. A next study should prespecify the primary channel and score, preserve sensor-placement and calibration metadata, retain a participant-flow log, define exclusions before analysis, estimate reliability, and validate embeddings against held-out sessions or external outcomes.

7 Limitations

The analytic sample was five of eleven participants and depended on local data availability. Missing raw annex objects and absent or misaligned recordings may introduce selection bias. Sampling at 1 Hz limits temporal interpretation. Channel choice was not preregistered, sensor metadata were incomplete, and no negative-control stimulus set or independent validation cohort was available. Semantic association responses and rubber-hand-illusion data were not analyzed. Statistics were exploratory and no adjustment for multiple analytic choices was applied. The Stage 2 framework is not itself a preregistration: exact endpoints, thresholds, sample size, intervention, and governance require investigator and ethics approval before being frozen. The current repository is not publicly accessible and the deidentified derived dataset is not yet deposited in a durable public archive.

8 Data and code availability

The analysis code, event-processing logic, and manuscript sources are maintained in the project repository. At the time of manuscript preparation the repository is private and participant-level recordings are not publicly released. A deidentified derived table, versioned code snapshot, environment specification, and checksums have not yet been deposited in a durable archive; the present analysis should therefore not be described as independently reproducible. Access to underlying research data is subject to ethics approval, consent, and an appropriate data-use process; inquiries may be directed to the corresponding author.

9 Earlier public version

An earlier, non-peer-reviewed project website article and PDF presented a broader thermodynamic interpretation and stronger claims about identified structures. This manuscript substantially revises those claims: no physical energy quantity, Jungian complex, or collective-unconscious structure was validated by the pilot. The earlier version should be disclosed to the receiving journal as related public material and identified as superseded by this analysis. See <https://spirit-in-physics.org/paper>.

10 Conclusion

This manuscript combines a Stage 1 feasibility result with a prospective Stage 2 validation framework for a physical hypothesis of spirit. Physical realization, response energy, Jungian complexes, and collective unconscious are explicit targets in a gated evidence program. The present data establish Level 0 constructibility; preregistered reliability, held-out prediction, controlled perturbation, energetic calibration, independent construct validity, and replication determine whether the hypotheses advance.

Author contributions

The contribution statement and final author list are provisional. Each named author must confirm their own role and approve the final text before submission.

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